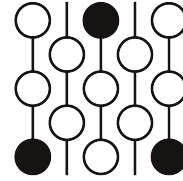


Review of Submission 5A

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. The AI proves the existence and uniqueness of the invariant measure for the given SPDE. The existence is proved by taking the limit of the invariant measures of the approximate equations. The uniqueness is derived from the fact that the transition function of the solution to the SPDE is absolutely continuous with respect to the invariant measure. The latter is proved using a version of Girsanov's theorem, which requires an L^2 bound on the drift term in the equation. The key part of the argument is a "uniform estimate on the regularized drift," which is used to obtain the aforementioned L^2 bound and is proved using a stochastic sewing lemma.

Novelty. The existence proof is standard and does not contain any novelty. To prove uniqueness, the AI uses a clever argument involving a stochastic sewing lemma. While this lemma itself is not novel, its application to the given problem is interesting and may be considered a novelty. Moreover, applying the lemma requires some analysis, which was carried out by the AI.

Exposition. The referee has not found any major mistakes in the solution, but some editing is needed to make the exposition and computations more transparent. Providing precise references would also improve the presentation.

2 Recommendation

The solution is essentially flawless. The mathematical content is correct, although some additional explanations and minor edits would improve the exposition.

3 Submission with inline comments

Uniqueness of the invariant measure

Problem statement and interpretation

Put $E = C_0([0, 1])$, $H = L^2(0, 1)$, and let S_t be the Dirichlet heat semigroup generated by $\frac{1}{2}\Delta$. I use the Dirichlet version of the Athreya–Butkovsky–Lê–Mytnik approximation assumption in the following form. For each deterministic initial condition $x \in E$, if $b_n = G_{1/n}$, then the regularized equation has the mild solution

$$u_t^n = S_t x + V_t + K_t^n, \quad K_t^n = \int_0^t S_{t-r} b_n(u_r^n) dr, \quad V_t = \int_0^t S_{t-r} dW_r,$$

and, for every $t > 0$ and every $\Phi \in C_b(E)$,

$$P_t^n \Phi(x) := \mathbb{E} \Phi(u_t^n) \longrightarrow P_t \Phi(x). \quad (1)$$

Thus I interpret the stated weak convergence of $u^n(t)$ to $u(t)$ as holding for all deterministic starting points; this is the form needed for a statement about invariant measures of the whole Markov semigroup.

Lemma 1 (Uniform estimate on the regularized drift). *For each $T < \infty$ there is $C_T < \infty$, independent of n and $x \in E$, such that*

$$\|K_t^n - S_{t-s} K_s^n\|_{L^2(\Omega; H)} \leq C_T |t - s|^{3/4}, \quad 0 \leq s < t \leq T. \quad (2)$$

Proof. Let p_t be the Dirichlet kernel and $d(y) = y \wedge (1 - y)$. For $0 < a \leq 2T$, the reflection formula for the interval and comparison with the two half-lines at the nearer boundary give

$$p_a(z, y) \leq C a^{-1/2} e^{-|z-y|^2/(Ca)}, \quad p_a(y, y) \asymp a^{-1/2} (1 \wedge d(y)^2/a). \quad (3)$$

For example, if $d(y) \leq \sqrt{a}$, the half-line expression $(2\pi a)^{-1/2} (1 - e^{-2d(y)^2/a})$ is comparable to $d(y)^2 a^{-3/2}$; if $d(y) \geq \sqrt{a}$, the reflected terms are exponentially smaller than the free diagonal term. The finitely many remaining images are absorbed in a constant depending only on T .

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A proof of these bounds could be more explicit.

Hence, for $\rho_\tau(y) = \int_0^\tau p_{2r}(y, y) dr$,

$$\rho_\tau(y) \asymp \sqrt{\tau} \wedge d(y), \quad \int_0^1 \rho_\tau(y)^{-1/2} dy \leq C_T \tau^{-1/4}. \quad (4)$$

Indeed the integral of $r^{-1/2} (1 \wedge d^2/r)$ is comparable to $\sqrt{\tau}$ when $d \geq \sqrt{\tau}$, and to d otherwise. Also

$$\sup_z \int_0^1 p_a(z, y) \rho_\tau(y)^{-1/2} dy \leq C_T (a^{-1/4} + \tau^{-1/4}). \quad (5)$$

To see this, use $\rho_\tau^{-1/2} \leq C(\tau^{-1/4} + d^{-1/2})$. The first part integrates to $C\tau^{-1/4}$. For the second, bound $d(y)^{-1/2} \leq y^{-1/2} + (1-y)^{-1/2}$ and use $y = \sqrt{a}r$ near the left boundary (similarly near the right) to get $C a^{-1/4} \sup_{\alpha \geq 0} \int_0^\infty e^{-(r-\alpha)^2/C} r^{-1/2} dr < \infty$. Finally, for $0 < \tau \leq T$,

$$\|\rho_\tau^{-1} S_\tau f\|_2 \leq C_T \tau^{-1/2} \|f\|_2. \quad (6)$$

Here $\rho_\tau^{-1} \leq C(\tau^{-1/2} + d^{-1})$; $S_\tau f \in H_0^1$; Hardy's inequality gives $\|S_\tau f/d\|_2 \leq C\|\partial_x S_\tau f\|_2$; and $\|\partial_x S_\tau f\|_2 \leq C\tau^{-1/2}\|f\|_2$.

Write $D_{s,t}^n = K_t^n - S_{t-s}K_s^n$. Freeze the equation at time s :

$$A_{s,t}^n = \int_s^t S_{t-r}b_n(S_{r-s}u_s^n + V_r - S_{r-s}V_s) dr. \quad (7)$$

For fixed n , if Π is a partition of $[s, t]$, then

$$\sum_{[a,b] \in \Pi} S_{t-b}A_{a,b}^n \rightarrow D_{s,t}^n \quad \text{in } L^2(H). \quad (8)$$

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The mesh size goes to zero in this limit, which is not stated.

Indeed the summand differs from

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writing a formula more explicitly would improve the argument

$\int_a^b S_{b-r}b_n(u_r^n) dr$ by at most $C_n(b-a)^2$ in $L^2(H)$, because b_n is Lipschitz and the two arguments differ only by the drift accumulated on $[a, r]$, whose H -norm is at most $C_n(r-a)$. After applying S_{t-b} and summing, the error is $O_n((t-s)|\Pi|)$.

The estimates below are uniform in n . Given $\mathcal{F}_s$,

$$Z_r(y) := S_{r-s}u_s^n(y) + V_r(y) - S_{r-s}V_s(y)$$

is Gaussian with variance $\rho_{r-s}(y)$. Since b_n is a probability density, $\sup_m \mathbb{E}b_n(m + N(0, \sigma^2)) \leq C\sigma^{-1}$. Moreover, given $\mathcal{F}_r$, the future part of $Z_q(z)$, $q > r$, has variance

$$\int_r^q \int_0^1 p_{q-\ell}(z, \xi)^2 d\xi d\ell = \int_0^{q-r} p_{2a}(z, z) da = \rho_{q-r}(z).$$

Thus, for $s < r < q$,

$$\mathbb{E}_s b_n(Z_r(y))b_n(Z_q(z)) \leq C\rho_{r-s}(y)^{-1/2}\rho_{q-r}(z)^{-1/2}. \quad (9)$$

Using $\int p_{t-r}(x, y)p_{t-q}(x, z)dx = p_{2t-r-q}(y, z)$,

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what is the integration domain?

then (5) in y and (4) in z , we obtain

$$\mathbb{E}_s \|A_{s,t}^n\|_2^2 \leq C \int_s^t \int_r^t \left((2t-r-q)^{-1/4} + (r-s)^{-1/4} \right) (q-r)^{-1/4} dq dr \leq C|t-s|^{3/2}. \quad (10)$$

The last inequality is just scaling on the triangle $0 \leq r-s \leq q-s \leq t-s$. Hence $\|A_{s,t}^n\|_{L^2 H} \leq C|t-s|^{3/4}$.

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The AI is using two L^2 norms, which have not been defined explicitly. This made reading the proof a bit complicated.

For $s < u < t$, let $\delta A_{s,u,t}^n = A_{s,t}^n - S_{t-u}A_{s,u}^n - A_{u,t}^n$. The pieces before u cancel. For $r > u$, the two frozen arguments differ by $S_{r-u}D_{s,u}^n$. Conditional on $\mathcal{F}_u$, the remaining noise at space point y has variance $\rho_{r-u}(y)$, and $\|(b_n * \varphi_\sigma)'\|_\infty \leq C\sigma^{-2}$, uniformly in n .

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φ_σ is undefined, although one can easily guess what it is

Therefore

$$|\mathbb{E}_u\{b_n(Z_r(y)) - b_n(Z_r(y) + h(y))\}| \leq C\rho_{r-u}(y)^{-1}|h(y)|.$$

Reviewer 1

The absolute value should be inside the expectation

With $h = S_{r-u}D_{s,u}^n$, (6) and contraction of S_t imply

$$\|\mathbb{E}_u\delta A_{s,u,t}^n\|_2 \leq C(t-u)^{1/2}\|D_{s,u}^n\|_2. \quad (11)$$

Taking $\mathbb{E}_s$ and using Jensen gives the same estimate with $\mathbb{E}_s$ in place of $\mathbb{E}_u$.

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One does not get the same estimate: There should be the expectation on the right-hand side.

We need only the following dyadic sewing fact. Suppose $A_{s,t}$ is H -valued and $\mathcal{F}_t$ -measurable, and for every dyadic interval $[a, b]$ with midpoint u ,

$$\|\mathbb{E}_a\delta A_{a,u,b}\|_{L^2H} \leq \Gamma_1(b-a)^{1+\varepsilon_1}, \quad \|\delta A_{a,u,b} - \mathbb{E}_a\delta A_{a,u,b}\|_{L^2H} \leq \Gamma_2(b-a)^{1/2+\varepsilon_2}.$$

If $J_m = \sum_i S_{t-t_{i+1}}A_{t_i,t_{i+1}}$ over the 2^m equal subintervals of $[s, t]$, then J_m converges

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Converges in which sense?

and

$$\|\mathcal{A}_{s,t} - A_{s,t}\|_{L^2H} \leq C(\Gamma_1|t-s|^{1+\varepsilon_1} + \Gamma_2|t-s|^{1/2+\varepsilon_2}). \quad (12)$$

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$\mathcal{A}$ is undefined

Indeed, $J_{m+1} - J_m$ is the sum of $-S_{t-b}\delta A_{a,u,b}$. The conditional-mean parts are bounded by the triangle inequality, giving a factor $2^m(|t-s|2^{-m})^{1+\varepsilon_1}$. The centered parts are orthogonal: for two level- m intervals with the first lying earlier, the earlier summand is $\mathcal{F}_a$ -measurable for the later one, while the later summand has conditional mean zero given $\mathcal{F}_a$. Thus the square norm is the sum of the square norms. Summing the resulting geometric series proves (12).

For fixed n , $Y_n(h) = \sup_{0 < t-s \leq h} \|D_{s,t}^n\|_{L^2H}/|t-s|^{3/4}$

h means two different things in this proof

is finite, because b_n is bounded. On intervals of length at most h , (11) gives

$$\|\mathbb{E}_a \delta A_{a,u,b}^n\|_{L^2 H} \leq C Y_n(h) (b-a)^{5/4}.$$

One has $b-a \leq h$ in this bound

Also

$$\|\delta A_{a,u,b}^n - \mathbb{E}_a \delta A_{a,u,b}^n\|_{L^2 H} \leq 2 \|\delta A_{a,u,b}^n\|_{L^2 H} \leq C (b-a)^{3/4},$$

because δA is a sum of three A -terms and (10) applies to each. The dyadic sewing limit is $D_{s,t}^n$ by (8). Combining (10) and (12),

$$\|D_{s,t}^n\|_{L^2 H} \leq C |t-s|^{3/4} + C Y_n(h) |t-s|^{5/4}.$$

Thus $Y_n(h) \leq C + C h^{1/2} Y_n(h)$. Choose $h_0 > 0$, depending only on T , so that $C h_0^{1/2} \leq 1/2$; then $Y_n(h_0) \leq 2C$, uniformly in n . For longer intervals use the additivity $D_{s,t}^n = S_{t-r} D_{s,r}^n + D_{r,t}^n$ and split into pieces of length at most h_0 . This gives a linear bound $C_T |t-s|$, which is bounded by $C_T |t-s|^{3/4}$ on $0 \leq t-s \leq T$. The lemma follows. $\square$

The reversible reference measure and its gap

Let γ be Brownian bridge measure on E , the centered Gaussian measure with covariance $(-\Delta_D)^{-1}$. Define

$$\Lambda(v) = \int_0^1 \mathbf{1}_{\{v(r) > 0\}} dr, \quad \pi(dv) = Z^{-1} e^{2\Lambda(v)} \gamma(dv). \quad (13)$$

For $B'_n = b_n$, $B_n(a) = \int_{-\infty}^a b_n(r) dr$, put

$$\pi_n(dv) = Z_n^{-1} \exp \left(2 \int_0^1 B_n(v(r)) dr \right) \gamma(dv).$$

Since $0 \leq B_n \leq 1$, $B_n(a) \rightarrow \mathbf{1}_{\{a > 0\}}$ for $a \neq 0$, and a Brownian bridge spends zero Lebesgue time at a fixed level,

$$\|\pi_n - \pi\|_{TV} \rightarrow 0. \quad (14)$$

All densities $d\pi_n/d\gamma$ and $d\pi/d\gamma$ are bounded above and below by deterministic positive constants.

Lemma 2. *There are constants $C, c > 0$, independent of n , such that for every bounded continuous f, g and every $t \geq 0$,*

$$\int P_t^n f d\pi_n = \int f d\pi_n, \quad \int g P_t^n f d\pi_n = \int f P_t^n g d\pi_n, \quad (15)$$

and

$$\text{Var}_{\pi_n}(P_t^n f) \leq e^{-ct} \text{Var}_{\pi_n}(f). \quad (16)$$

Proof. Let Π_m be projection onto the first m Dirichlet modes and set

$$U_{n,m}(v) = \int_0^1 B_n(\Pi_m v(r)) dr.$$

Its H -gradient is $\nabla_H U_{n,m}(v) = \Pi_m b_n(\Pi_m v)$. Let $P_t^{n,m}$ be the semigroup of

$$u_t^{n,m} = S_t x + V_t + \int_0^t S_{t-r} \Pi_m b_n(\Pi_m u_r^{n,m}) dr.$$

The first m Fourier coordinates solve the finite-dimensional Langevin diffusion

$$dX_j = -(\lambda_j/2)X_j dt + \partial_j U_{n,m}(X) dt + d\beta_j, \quad j \leq m,$$

λ_j is defined much later

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and the tail is an independent OU process. Hence, by finite-dimensional integration by parts and the standard OU integration by parts in the tail, $P_t^{n,m}$ is reversible with respect to $\pi_{n,m} = Z_{n,m}^{-1} e^{2U_{n,m}\gamma}$, and its Dirichlet form is $\mathcal{E}_{n,m}(F, F) = \frac{1}{2} \int \|\nabla_H F\|_2^2 d\pi_{n,m}$ on smooth cylinder functions.

The measures $\pi_{n,m}$ converge to π_n in total variation: $\Pi_m v \rightarrow v$ in H , γ -a.s., and B_n is bounded and Lipschitz. For fixed n , $u_t^{n,m} \rightarrow u_t^n$ in probability in E . In H , this follows from Gronwall, since $\Pi_m b_n(\Pi_m z)$ is uniformly Lipschitz in z for fixed n , and for each H -valued y , $\Pi_m b_n(\Pi_m y) \rightarrow b_n(y)$ in H , dominated by $2\|b_n\|_\infty$. For convergence in E at a fixed $t > 0$, split the drift convolution into $[0, t - \varepsilon]$ and $[t - \varepsilon, t]$. On the first part, $S_{t-r} : H \rightarrow H_0^\alpha \subset E$, $\alpha > 1/2$, is bounded. On the short terminal part,

$$\int_{t-\varepsilon}^t \|S_{t-r} F_r\|_\infty dr \leq C \|b_n\|_\infty \int_0^\varepsilon a^{-1/4} da,$$

for both $F_r = \Pi_m b_n(\Pi_m u_r^{n,m})$ and $F_r = b_n(u_r^n)$, because Π_m is an L^2 -contraction and $S_a : H \rightarrow E$ has norm $C a^{-1/4}$. This tends to zero with ε , uniformly in m .

Consequently the reversible identities pass to $m = \infty$. For instance, if $h_m = d\pi_{n,m}/d\gamma$, then $h_m \rightarrow h_n = d\pi_n/d\gamma$ in $L^1(\gamma)$ and the h_m 's are uniformly bounded; together with $P_t^{n,m} f(x) \rightarrow P_t^n f(x)$ and bounded convergence this gives (15).

It remains to record the gap uniformly. The Gaussian Poincare inequality for γ in H -gradient form is

$$\text{Var}_\gamma(F) \leq \lambda_1^{-1} \int \|\nabla_H F\|_2^2 d\gamma$$

on smooth cylinder functions. Since all densities $d\pi_{n,m}/d\gamma$ lie in a fixed interval $[a_0, a_0^{-1}]$, the Holley–Stroock bounded perturbation argument gives

$$\text{Var}_{\pi_{n,m}}(F) \leq C \mathcal{E}_{n,m}(F, F),$$

with C independent of n, m . For a symmetric Markov semigroup this Poincare inequality is equivalent to exponential L^2 -decay, so

$$\text{Var}_{\pi_{n,m}}(P_t^{n,m} f) \leq e^{-ct} \text{Var}_{\pi_{n,m}}(f).$$

Letting $m \rightarrow \infty$, using the convergence just proved and the uniformly bounded densities, yields (16). $\square$

Proposition 1. *The measure π is P_t -invariant, P_t is symmetric on $L^2(\pi)$, and*

$$\text{Var}_\pi(P_t f) \leq e^{-ct} \text{Var}_\pi(f), \quad f \in L^2(\pi), \quad t \geq 0. \quad (17)$$

Proof. For $f, g \in C_b(E)$, (1), (14), (15), and the uniform boundedness of the densities imply, by dominated convergence with respect to γ ,

$$\int P_t f d\pi = \lim_n \int P_t^n f d\pi_n = \int f d\pi, \quad \int g P_t f d\pi = \int f P_t g d\pi.$$

On the Polish space E , these identities identify the measure $\int \pi(dx) P_t(x, \cdot)$ with π , and the measure $\pi(dx) P_t(x, dy)$ with its transpose; hence invariance and detailed balance hold for all bounded Borel functions. Similarly, (16) passes to the limit for bounded continuous f . Finally $C_b(E)$ is dense in $L^2(\pi)$, and an invariant Markov operator is an L^2 -contraction, so symmetry and (17) extend to all $L^2(\pi)$. $\square$

Fixed-time absolute continuity

Lemma 3. *For every deterministic $x \in E$ and every $t > 0$,*

$$P_t(x, \cdot) \ll \gamma,$$

and hence $P_t(x, \cdot) \ll \pi$.

Proof. Let $\nu_t^x = \text{Law}(S_t x + V_t)$. We first bound $\text{Ent}(P_t^n(x, \cdot) \mid \nu_t^x)$ uniformly in n . Put $r_k = t(1 - 2^{-k})$, $\Delta_k = r_{k+1} - r_k$, and $D_k^n = K_{r_{k+1}}^n - S_{\Delta_k} K_{r_k}^n$. Define

$$F_s^n = \sum_{k \geq 0} \frac{\mathbf{1}_{(r_{k+1}, r_{k+2}]}(s)}{\Delta_{k+1}} S_{s-r_{k+1}} D_k^n. \quad (18)$$

The process is predictable because D_k^n is $\mathcal{F}_{r_{k+1}}$ -measurable. Lemma 1 and contraction of S_t give

$$\mathbb{E} \int_0^t \|F_s^n\|_2^2 ds \leq C_t \sum_{k \geq 0} \frac{\Delta_k^{3/2}}{\Delta_{k+1}} < \infty, \quad (19)$$

uniformly in n . For partial sums,

$$\int_0^t S_{t-s} F_s^{n,N} ds = \sum_{k=0}^N S_{t-r_{k+1}} D_k^n = S_{t-r_{N+1}} K_{r_{N+1}}^n,$$

where $F^{n,N}$ denotes the sum in (18) up to $k = N$. Since $r_{N+1} \uparrow t$, the regularized mild solution gives K^n continuous with values in E , and $S_a \rightarrow I$ strongly on E ,

$$\int_0^t S_{t-s} F_s^n ds = K_t^n \quad (20)$$

in probability in E (and hence in H). The state space assertion is legitimate because, for every deterministic $F \in L^2(0, t; H)$,

$$\sum_j \lambda_j^\alpha \left| \int_0^t e^{-\lambda_j(t-s)/2} F_j(s) ds \right|^2 \leq \sum_j \lambda_j^{\alpha-1} \int_0^t |F_j(s)|^2 ds \leq C \|F\|_{L^2(0,t;H)}^2$$

for $0 < \alpha < 1$; choosing $\alpha > 1/2$ gives an element of $H_0^\alpha(0, 1) \subset E$.

We use the following standard entropy form of Girsanov, and recall its proof. If F is predictable, H -valued, and has bounded energy, then

$$X^F = S_t x + V_t + \int_0^t S_{t-s} F_s ds$$

satisfies

$$\text{Ent}(\text{Law}(X^F) \mid \nu_t^x) \leq \frac{1}{2} \mathbb{E} \int_0^t \|F_s\|_2^2 ds. \quad (21)$$

For the first M Brownian coordinates this is finite-dimensional Girsanov: under the tilted measure, the shifted coordinates $\beta_j + \int_0^t F_j(s) ds$, $j \leq M$, are Brownian, and the entropy cost is $\frac{1}{2} \mathbb{E} \int_0^t \sum_{j \leq M} F_j(s)^2 ds$. Applying data processing to the deterministic-kernel stochastic convolution gives the same bound for the first M terminal Fourier coordinates. Since the Borel sigma-field of E is generated by these coordinates and relative entropy is the supremum over increasing finite-coordinate sigma-fields, (21) follows.

Localize F^n by $\tau_R = \inf\{s : \int_0^s \|F_r^n\|_2^2 dr > R\}$ and $\hat{F}^{n,R} = F^n \mathbf{1}_{[0, \tau_R]}$. Let

$$X_R = S_t x + V_t + \int_0^t S_{t-s} \hat{F}_s^{n,R} ds.$$

Then (21) and (19) give

$$\text{Ent}(\text{Law}(X_R) \mid \nu_t^x) \leq \frac{1}{2} \mathbb{E} \int_0^t \|F_s^n\|_2^2 ds. \quad (22)$$

On $\{\tau_R > t\}$, (20) gives $X_R = u_t^n$, while $\mathbb{P}(\tau_R \leq t) \leq R^{-1} \mathbb{E} \int_0^t \|F_s^n\|_2^2 ds \rightarrow 0$. Hence $X_R \rightarrow u_t^n$ in probability in E . Lower semicontinuity of relative entropy on the Polish space E , then (19) and (22), yield

$$\sup_n \text{Ent}(P_t^n(x, \cdot) \mid \nu_t^x) < \infty. \quad (23)$$

By the assumed weak convergence (1) and lower semicontinuity again, $\text{Ent}(P_t(x, \cdot) \mid \nu_t^x) < \infty$. Thus $P_t(x, \cdot) \ll \nu_t^x$.

It remains only to compare Gaussian measures. In the basis $e_j(r) = \sqrt{2} \sin(j\pi r)$, $\lambda_j = j^2 \pi^2$, the covariance of ν_t^x has eigenvalues

$$q_j(t) = \frac{1 - e^{-\lambda_j t}}{\lambda_j},$$

whereas γ has eigenvalues λ_j^{-1} . The ratios $1 - e^{-\lambda_j t}$ are bounded away from zero and differ from 1 by a square-summable sequence. The Cameron–Martin spaces are therefore both $H_0^1(0, 1)$, and $S_t x \in H_0^1$. Feldman–Hajek gives $\nu_t^x \sim \gamma$. $\square$

Conclusion

Let η be an invariant probability measure. If $\pi(A) = 0$, then Lemma 3 gives $P_t(y, A) = 0$ for every $y \in E$, and hence

$$\eta(A) = \int_E P_t(y, A) \eta(dy) = 0.$$

Thus $\eta \ll \pi$; write $d\eta = h d\pi$. Detailed balance gives the L^1 -duality

$$\int f P_t g d\pi = \int P_t f g d\pi$$

for bounded Borel f and $g \in L^1(\pi)$, by truncating g .

what exactly does "truncating of g " mean?

Reviewer 1

Therefore

$$\int f h d\pi = \int P_t f d\eta = \int P_t f h d\pi = \int f P_t h d\pi,$$

so $P_t h = h$ in $L^1(\pi)$. For $a > 0$, let $h_a = h \wedge a$. Since $r \mapsto r \wedge a$ is concave,

$$P_t h_a \leq (P_t h) \wedge a = h_a.$$

The two sides have the same π -integral; therefore $P_t h_a = h_a$. Now $h_a \in L^2(\pi)$, and the spectral gap (17) implies $\text{Var}_\pi(h_a) = 0$. Thus h_a is constant for every a . Letting $a \uparrow \infty$ and using $\int h d\pi = 1$, we obtain $h = 1$ π -a.s. Hence $\eta = \pi$.

Therefore P_t has at most one invariant probability measure on $C_0([0, 1])$. In fact, if an invariant probability exists, it is the Gibbs measure π in (13).

References

- [1] S. Athreya, O. Butkovsky, K. Lê and L. Mytnik, *Well-posedness of stochastic heat equation with distributional drift and skew stochastic heat equation*, Comm. Pure Appl. Math. 77 (2024), 2708–2777.
- [2] K. Lê, *Stochastic sewing in Banach spaces*, Electron. J. Probab. 28 (2023), paper no. 26, 1–22.
- [3] K. Lê, *A stochastic sewing lemma and applications*, Electron. J. Probab. 25 (2020), paper no. 38, 1–55.
- [4] V. I. Bogachev, *Gaussian Measures*, American Mathematical Society, 1998.
- [5] G. Da Prato and J. Zabczyk, *Ergodicity for Infinite Dimensional Systems*, Cambridge University Press, 1996.
- [6] Z.-M. Ma and M. Röckner, *Introduction to the Theory of (Non-Symmetric) Dirichlet Forms*, Springer, 1992.

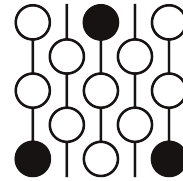
- [7] S. K. Bounebacha and L. Zambotti, *A skew stochastic heat equation*, J. Theoret. Probab. 27 (2014), 168–201.

Review of Submission 5A

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. We are confident that the submission gives a correct proof of the problem. The key step of the submission's argument is Lemma 1, which uses the stochastic sewing lemma to prove an estimate on the drift. This is combined with a Girsanov-type entropy bound, which shows that, for any fixed $t > 0$ and any initial condition x , the transition probability $P_t(x, \bullet)$ is absolutely continuous with respect to the Brownian bridge measure γ . Moreover, an explicit invariant measure μ for the dynamic can be exhibited, and it is easy to see μ and γ are equivalent, so $P_t(x, \bullet)$ must also be absolutely continuous with respect to μ .

The most important technical ingredient is the drift estimate. This is proven using the stochastic sewing lemma, which controls the drift using carefully chosen time splitting approximations whose individual terms satisfy suitable probabilistic estimates with respect to the filtration generated by the noise.

Novelty. The key technical ingredient, that is the use of the stochastic sewing lemma to estimate the drift, is an important novelty. In fact, the human solution also makes use of the stochastic sewing lemma (in a different way) and the accompanying novelty statement identifies this as an important novelty in the human solution. The ergodic theory argument which the submission used to prove the desired result is quite classical - the idea is that the drift is a "small" perturbation so one has absolute continuity with respect to the unique invariant measure of the linear dynamic.

Proving the estimate on the drift needed to make this argument go through is far from clear, and that is where the implementation of stochastic sewing lemma does the heavy lifting.

Exposition. Since the submission is correct, we also make remarks on exposition. For the most part, the exposition in the submission is fairly clear.

The main stylistic complaint we would make regarding exposition is that the submission does not make detailed inline citations - it only cites works in the bibliography. However, it does look like the bibliography contains all key works that should be cited for the submission. Additional issues are that the exposition becomes a bit terse and implicit in a few places and there are also places with undefined notation.

In comparison with the human solution, this submission is more streamlined and conceptually simpler. Moreover, the key intermediate result this submission proves appears to be stronger than

the intermediate results proven in the human solution.

2 Recommendation

On the basis of the correctness and novelty of the submission, we would categorize the submission as **essentially flawless**.

3 Submission with inline comments

Uniqueness of the invariant measure

Problem statement and interpretation

Put $E = C_0([0, 1])$, $H = L^2(0, 1)$, and let S_t be the Dirichlet heat semigroup generated by $\frac{1}{2}\Delta$. I use the Dirichlet version of the Athreya–Butkovsky–Lê–Mytnik approximation assumption in the following form. For each deterministic initial condition $x \in E$, if $b_n = G_{1/n}$, then the regularized equation has the mild solution

$$u_t^n = S_t x + V_t + K_t^n, \quad K_t^n = \int_0^t S_{t-r} b_n(u_r^n) dr, \quad V_t = \int_0^t S_{t-r} dW_r,$$

and, for every $t > 0$ and every $\Phi \in C_b(E)$,

$$P_t^n \Phi(x) := \mathbb{E} \Phi(u_t^n) \longrightarrow P_t \Phi(x). \quad (1)$$

Thus I interpret the stated weak convergence of $u^n(t)$ to $u(t)$ as holding for all deterministic starting points; this is the form needed for a statement about invariant measures of the whole Markov semigroup.

Lemma 1 (Uniform estimate on the regularized drift). *For each $T < \infty$ there is $C_T < \infty$, independent of n and $x \in E$, such that*

$$\|K_t^n - S_{t-s} K_s^n\|_{L^2(\Omega; H)} \leq C_T |t - s|^{3/4}, \quad 0 \leq s < t \leq T. \quad (2)$$

Proof. Let p_t be the Dirichlet kernel and $d(y) = y \wedge (1 - y)$. For $0 < a \leq 2T$, the reflection formula for the interval and comparison with the two half-lines at the nearer boundary give

$$p_a(z, y) \leq C a^{-1/2} e^{-|z-y|^2/(Ca)}, \quad p_a(y, y) \asymp a^{-1/2} (1 \wedge d(y)^2/a). \quad (3)$$

For example, if $d(y) \leq \sqrt{a}$, the half-line expression $(2\pi a)^{-1/2} (1 - e^{-2d(y)^2/a})$ is comparable to $d(y)^2 a^{-3/2}$; if $d(y) \geq \sqrt{a}$, the reflected terms are exponentially smaller than the free diagonal term. The finitely many remaining images are absorbed in a constant depending only on T .

Stylistic - The exposition here is a bit terse and implicit, presenting some formulae or citing a two-sided bound from the literature would make things clearer.

Hence, for $\rho_\tau(y) = \int_0^\tau p_{2r}(y, y) dr$,

$$\rho_\tau(y) \asymp \sqrt{\tau} \wedge d(y), \quad \int_0^1 \rho_\tau(y)^{-1/2} dy \leq C_T \tau^{-1/4}. \quad (4)$$

Indeed the integral of $r^{-1/2}(1 \wedge d^2/r)$ is comparable to $\sqrt{\tau}$ when $d \geq \sqrt{\tau}$, and to d otherwise. Also

$$\sup_z \int_0^1 p_a(z, y) \rho_\tau(y)^{-1/2} dy \leq C_T(a^{-1/4} + \tau^{-1/4}). \quad (5)$$

To see this, use $\rho_\tau^{-1/2} \leq C(\tau^{-1/4} + d^{-1/2})$. The first part integrates to $C\tau^{-1/4}$. For the second, bound $d(y)^{-1/2} \leq y^{-1/2} + (1-y)^{-1/2}$ and use $y = \sqrt{a}r$ near the left boundary (similarly near the right) to get $Ca^{-1/4} \sup_{\alpha \geq 0} \int_0^\infty e^{-(r-\alpha)^2/C} r^{-1/2} dr < \infty$. Finally, for $0 < \tau \leq T$,

$$\|\rho_\tau^{-1} S_\tau f\|_2 \leq C_T \tau^{-1/2} \|f\|_2. \quad (6)$$

Here $\rho_\tau^{-1} \leq C(\tau^{-1/2} + d^{-1})$; $S_\tau f \in H_0^1$; Hardy's inequality gives $\|S_\tau f/d\|_2 \leq C\|\partial_x S_\tau f\|_2$; and $\|\partial_x S_\tau f\|_2 \leq C\tau^{-1/2}\|f\|_2$.

Write $D_{s,t}^n = K_t^n - S_{t-s}K_s^n$. Freeze the equation at time s :

$$A_{s,t}^n = \int_s^t S_{t-r} b_n(S_{r-s} u_s^n + V_r - S_{r-s} V_s) dr. \quad (7)$$

For fixed n , if Π is a partition of $[s, t]$, then

$$\sum_{[a,b] \in \Pi} S_{t-b} A_{a,b}^n \rightarrow D_{s,t}^n \quad \text{in } L^2(H). \quad (8)$$

Indeed the summand differs from $\int_a^b S_{b-r} b_n(u_r^n) dr$ by at most $C_n(b-a)^2$ in $L^2(H)$, because b_n is Lipschitz and the two arguments differ only by the drift accumulated on $[a, r]$, whose H -norm is at most $C_n(r-a)$. After applying S_{t-b} and summing, the error is $O_n((t-s)|\Pi|)$.

The estimates below are uniform in n . Given $\mathcal{F}_s$,

$$Z_r(y) := S_{r-s} u_s^n(y) + V_r(y) - S_{r-s} V_s(y)$$

is Gaussian with variance $\rho_{r-s}(y)$. Since b_n is a probability density, $\sup_m \mathbb{E} b_n(m + N(0, \sigma^2)) \leq C\sigma^{-1}$. Moreover, given $\mathcal{F}_r$, the future part of $Z_q(z)$, $q > r$, has variance

$$\int_r^q \int_0^1 p_{q-\ell}(z, \xi)^2 d\xi d\ell = \int_0^{q-r} p_{2a}(z, z) da = \rho_{q-r}(z).$$

Thus, for $s < r < q$,

$$\mathbb{E}_s b_n(Z_r(y)) b_n(Z_q(z)) \leq C \rho_{r-s}(y)^{-1/2} \rho_{q-r}(z)^{-1/2}. \quad (9)$$

Using $\int p_{t-r}(x, y)p_{t-q}(x, z)dx = p_{2t-r-q}(y, z)$, then (5) in y and (4) in z , we obtain

$$\mathbb{E}_s \|A_{s,t}^n\|_2^2 \leq C \int_s^t \int_r^t \left((2t-r-q)^{-1/4} + (r-s)^{-1/4} \right) (q-r)^{-1/4} dq dr \leq C|t-s|^{3/2}. \quad (10)$$

The last inequality is just scaling on the triangle $0 \leq r-s \leq q-s \leq t-s$. Hence $\|A_{s,t}^n\|_{L^2 H} \leq C|t-s|^{3/4}$.

For $s < u < t$, let $\delta A_{s,u,t}^n = A_{s,t}^n - S_{t-u}A_{s,u}^n - A_{u,t}^n$. The pieces before u cancel. For $r > u$, the two frozen arguments differ by $S_{r-u}D_{s,u}^n$. Conditional on $\mathcal{F}_u$, the remaining noise at space point y has variance $\rho_{r-u}(y)$, and $\|(b_n * \varphi_\sigma)'\|_\infty \leq C\sigma^{-2}$, uniformly in n .

Reviewer 2

Stylistic - φ_σ is not defined, but from context one can see it refers to the density of a $N(0, \sigma^2)$ random variable.

Therefore

$$|\mathbb{E}_u \{b_n(Z_r(y)) - b_n(Z_r(y) + h(y))\}| \leq C\rho_{r-u}(y)^{-1}|h(y)|.$$

With $h = S_{r-u}D_{s,u}^n$, (6) and contraction of S_t imply

$$\|\mathbb{E}_u \delta A_{s,u,t}^n\|_2 \leq C(t-u)^{1/2} \|D_{s,u}^n\|_2. \quad (11)$$

Taking $\mathbb{E}_s$ and using Jensen gives the same estimate with $\mathbb{E}_s$ in place of $\mathbb{E}_u$.

Reviewer 2

Stylistic - this last sentence is a bit unclear, as there would then also be an $\mathbb{E}_s$ on the right hand side (which is fine for the argument).

We need only the following dyadic sewing fact. Suppose $A_{s,t}$ is H -valued and $\mathcal{F}_t$ -measurable, and for every dyadic interval $[a, b]$ with midpoint u ,

$$\|\mathbb{E}_a \delta A_{a,u,b}\|_{L^2 H} \leq \Gamma_1(b-a)^{1+\varepsilon_1}, \quad \|\delta A_{a,u,b} - \mathbb{E}_a \delta A_{a,u,b}\|_{L^2 H} \leq \Gamma_2(b-a)^{1/2+\varepsilon_2}.$$

If $J_m = \sum_i S_{t-t_{i+1}} A_{t_i, t_{i+1}}$ over the 2^m equal subintervals of $[s, t]$, then J_m converges and

$$\|\mathcal{A}_{s,t} - A_{s,t}\|_{L^2 H} \leq C(\Gamma_1|t-s|^{1+\varepsilon_1} + \Gamma_2|t-s|^{1/2+\varepsilon_2}). \quad (12)$$

Reviewer 2

Stylistic - $\mathcal{A}_{s,t}$ is not defined, refers to the limit of ‘‘Riemann sum approximations’’, will be $D_{s,t}^n$.

Indeed, $J_{m+1} - J_m$ is the sum of $-S_{t-b}\delta A_{a,u,b}$. The conditional-mean parts are bounded by the triangle inequality, giving a factor $2^m(|t-s|2^{-m})^{1+\varepsilon_1}$. The centered parts are orthogonal: for two level- m intervals with the first lying earlier, the earlier summand is $\mathcal{F}_a$ -measurable for the later one, while the later summand has conditional mean zero given $\mathcal{F}_a$. Thus the square norm is the sum of the square norms. Summing the resulting geometric series proves (12).

For fixed n , $Y_n(h) = \sup_{0 < t-s \leq h} \|D_{s,t}^n\|_{L^2 H} / |t-s|^{3/4}$ is finite, because b_n is bounded. On intervals of length at most h , (11) gives

$$\|\mathbb{E}_a \delta A_{a,u,b}^n\|_{L^2 H} \leq CY_n(h)(b-a)^{5/4}.$$

Also

$$\|\delta A_{a,u,b}^n - \mathbb{E}_a \delta A_{a,u,b}^n\|_{L^2 H} \leq 2\|\delta A_{a,u,b}^n\|_{L^2 H} \leq C(b-a)^{3/4},$$

because δA is a sum of three A -terms and (10) applies to each. The dyadic sewing limit is $D_{s,t}^n$ by (8). Combining (10) and (12),

$$\|D_{s,t}^n\|_{L^2 H} \leq C|t-s|^{3/4} + CY_n(h)|t-s|^{5/4}.$$

Thus $Y_n(h) \leq C + Ch^{1/2}Y_n(h)$. Choose $h_0 > 0$, depending only on T , so that $Ch_0^{1/2} \leq 1/2$; then $Y_n(h_0) \leq 2C$, uniformly in n . For longer intervals use the additivity $D_{s,t}^n = S_{t-r}D_{s,r}^n + D_{r,t}^n$ and split into pieces of length at most h_0 . This gives a linear bound $C_T|t-s|$, which is bounded by $C_T|t-s|^{3/4}$ on $0 \leq t-s \leq T$. The lemma follows. $\square$

The reversible reference measure and its gap

Let γ be Brownian bridge measure on E , the centered Gaussian measure with covariance $(-\Delta_D)^{-1}$. Define

$$\Lambda(v) = \int_0^1 \mathbf{1}_{\{v(r)>0\}} dr, \quad \pi(dv) = Z^{-1} e^{2\Lambda(v)} \gamma(dv). \quad (13)$$

For $B'_n = b_n$, $B_n(a) = \int_{-\infty}^a b_n(r) dr$, put

$$\pi_n(dv) = Z_n^{-1} \exp\left(2 \int_0^1 B_n(v(r)) dr\right) \gamma(dv).$$

Since $0 \leq B_n \leq 1$, $B_n(a) \rightarrow \mathbf{1}_{\{a>0\}}$ for $a \neq 0$, and a Brownian bridge spends zero Lebesgue time at a fixed level,

$$\|\pi_n - \pi\|_{TV} \rightarrow 0. \quad (14)$$

All densities $d\pi_n/d\gamma$ and $d\pi/d\gamma$ are bounded above and below by deterministic positive constants.

Lemma 2. *There are constants $C, c > 0$, independent of n , such that for every bounded continuous f, g and every $t \geq 0$,*

$$\int P_t^n f d\pi_n = \int f d\pi_n, \quad \int g P_t^n f d\pi_n = \int f P_t^n g d\pi_n, \quad (15)$$

and

$$\text{Var}_{\pi_n}(P_t^n f) \leq e^{-ct} \text{Var}_{\pi_n}(f). \quad (16)$$

Proof. Let Π_m be projection onto the first m Dirichlet modes and set

$$U_{n,m}(v) = \int_0^1 B_n(\Pi_m v(r)) dr.$$

Its H -gradient is $\nabla_H U_{n,m}(v) = \Pi_m b_n(\Pi_m v)$. Let $P_t^{n,m}$ be the semigroup of

$$u_t^{n,m} = S_t x + V_t + \int_0^t S_{t-r} \Pi_m b_n(\Pi_m u_r^{n,m}) dr.$$

The first m Fourier coordinates solve the finite-dimensional Langevin diffusion

$$dX_j = -(\lambda_j/2)X_j dt + \partial_j U_{n,m}(X) dt + d\beta_j, \quad j \leq m,$$

and the tail is an independent OU process. Hence, by finite-dimensional integration by parts and the standard OU integration by parts in the tail, $P_t^{n,m}$ is reversible with respect to $\pi_{n,m} = Z_{n,m}^{-1} e^{2U_{n,m}} \gamma$, and its Dirichlet form is $\mathcal{E}_{n,m}(F, F) = \frac{1}{2} \int \|\nabla_H F\|_2^2 d\pi_{n,m}$ on smooth cylinder functions.

The measures $\pi_{n,m}$ converge to π_n in total variation: $\Pi_m v \rightarrow v$ in H , γ -a.s., and B_n is bounded and Lipschitz. For fixed n , $u_t^{n,m} \rightarrow u_t^n$ in probability in E . In H , this follows from Gronwall, since $\Pi_m b_n(\Pi_m z)$ is uniformly Lipschitz in z for fixed n , and for each H -valued y , $\Pi_m b_n(\Pi_m y) \rightarrow b_n(y)$ in H , dominated by $2\|b_n\|_\infty$. For convergence in E at a fixed $t > 0$, split the drift convolution into $[0, t - \varepsilon]$ and $[t - \varepsilon, t]$. On the first part, $S_{t-r} : H \rightarrow H_0^\alpha \subset E$, $\alpha > 1/2$, is bounded. On the short terminal part,

$$\int_{t-\varepsilon}^t \|S_{t-r} F_r\|_\infty dr \leq C \|b_n\|_\infty \int_0^\varepsilon a^{-1/4} da,$$

for both $F_r = \Pi_m b_n(\Pi_m u_r^{n,m})$ and $F_r = b_n(u_r^n)$, because Π_m is an L^2 -contraction and $S_a : H \rightarrow E$ has norm $C a^{-1/4}$. This tends to zero with ε , uniformly in m .

Consequently the reversible identities pass to $m = \infty$. For instance, if $h_m = d\pi_{n,m}/d\gamma$, then $h_m \rightarrow h_n = d\pi_n/d\gamma$ in $L^1(\gamma)$ and the h_m 's are uniformly bounded; together with $P_t^{n,m} f(x) \rightarrow P_t^n f(x)$ and bounded convergence this gives (15).

It remains to record the gap uniformly. The Gaussian Poincare inequality for γ in H -gradient form is

$$\text{Var}_\gamma(F) \leq \lambda_1^{-1} \int \|\nabla_H F\|_2^2 d\gamma$$

on smooth cylinder functions. Since all densities $d\pi_{n,m}/d\gamma$ lie in a fixed interval $[a_0, a_0^{-1}]$, the Holley–Stroock bounded perturbation argument gives

$$\text{Var}_{\pi_{n,m}}(F) \leq C \mathcal{E}_{n,m}(F, F),$$

with C independent of n, m . For a symmetric Markov semigroup this Poincare inequality is equivalent to exponential L^2 -decay, so

$$\text{Var}_{\pi_{n,m}}(P_t^{n,m} f) \leq e^{-ct} \text{Var}_{\pi_{n,m}}(f).$$

Letting $m \rightarrow \infty$, using the convergence just proved and the uniformly bounded densities, yields (16). $\square$

Proposition 1. *The measure π is P_t -invariant, P_t is symmetric on $L^2(\pi)$, and*

$$\text{Var}_\pi(P_t f) \leq e^{-ct} \text{Var}_\pi(f), \quad f \in L^2(\pi), \quad t \geq 0. \quad (17)$$

Proof. For $f, g \in C_b(E)$, (1), (14), (15), and the uniform boundedness of the densities imply, by dominated convergence with respect to γ ,

$$\int P_t f d\pi = \lim_n \int P_t^n f d\pi_n = \int f d\pi, \quad \int g P_t f d\pi = \int f P_t g d\pi.$$

On the Polish space E , these identities identify the measure $\int \pi(dx)P_t(x, \cdot)$ with π , and the measure $\pi(dx)P_t(x, dy)$ with its transpose; hence invariance and detailed balance hold for all bounded Borel functions. Similarly, (16) passes to the limit for bounded continuous f . Finally $C_b(E)$ is dense in $L^2(\pi)$, and an invariant Markov operator is an L^2 -contraction, so symmetry and (17) extend to all $L^2(\pi)$. $\square$

Fixed-time absolute continuity

Lemma 3. *For every deterministic $x \in E$ and every $t > 0$,*

$$P_t(x, \cdot) \ll \gamma,$$

and hence $P_t(x, \cdot) \ll \pi$.

Proof. Let $\nu_t^x = \text{Law}(S_t x + V_t)$. We first bound $\text{Ent}(P_t^n(x, \cdot) \mid \nu_t^x)$ uniformly in n . Put $r_k = t(1 - 2^{-k})$, $\Delta_k = r_{k+1} - r_k$, and $D_k^n = K_{r_{k+1}}^n - S_{\Delta_k} K_{r_k}^n$. Define

$$F_s^n = \sum_{k \geq 0} \frac{\mathbf{1}_{(r_{k+1}, r_{k+2}]}(s)}{\Delta_{k+1}} S_{s-r_{k+1}} D_k^n. \quad (18)$$

The process is predictable because D_k^n is $\mathcal{F}_{r_{k+1}}$ -measurable. Lemma 1 and contraction of S_t give

$$\mathbb{E} \int_0^t \|F_s^n\|_2^2 ds \leq C_t \sum_{k \geq 0} \frac{\Delta_k^{3/2}}{\Delta_{k+1}} < \infty, \quad (19)$$

uniformly in n . For partial sums,

$$\int_0^t S_{t-s} F_s^{n,N} ds = \sum_{k=0}^N S_{t-r_{k+1}} D_k^n = S_{t-r_{N+1}} K_{r_{N+1}}^n,$$

where $F^{n,N}$ denotes the sum in (18) up to $k = N$. Since $r_{N+1} \uparrow t$, the regularized mild solution gives K^n continuous with values in E , and $S_a \rightarrow I$ strongly on E ,

$$\int_0^t S_{t-s} F_s^n ds = K_t^n \quad (20)$$

in probability in E (and hence in H). The state space assertion is legitimate because, for every deterministic $F \in L^2(0, t; H)$,

$$\sum_j \lambda_j^\alpha \left| \int_0^t e^{-\lambda_j(t-s)/2} F_j(s) ds \right|^2 \leq \sum_j \lambda_j^{\alpha-1} \int_0^t |F_j(s)|^2 ds \leq C \|F\|_{L^2(0,t;H)}^2$$

for $0 < \alpha < 1$; choosing $\alpha > 1/2$ gives an element of $H_0^\alpha(0, 1) \subset E$.

We use the following standard entropy form of Girsanov, and recall its proof. If F is predictable, H -valued, and has bounded energy, then

$$X^F = S_t x + V_t + \int_0^t S_{t-s} F_s ds$$

satisfies

$$\text{Ent}(\text{Law}(X^F) \mid \nu_t^x) \leq \frac{1}{2} \mathbb{E} \int_0^t \|F_s\|_2^2 ds. \quad (21)$$

For the first M Brownian coordinates this is finite-dimensional Girsanov: under the tilted measure, the shifted coordinates $\beta_j + \int_0^\cdot F_j(s) ds$, $j \leq M$, are Brownian, and the entropy cost is $\frac{1}{2} \mathbb{E} \int_0^t \sum_{j \leq M} F_j(s)^2 ds$. Applying data processing to the deterministic-kernel stochastic convolution gives the same bound for the first M terminal Fourier coordinates. Since the Borel sigma-field of E is generated by these coordinates and relative entropy is the supremum over increasing finite-coordinate sigma-fields, (21) follows.

Reviewer 2

Stylistic - The exposition here is also a bit terse and implicit, writing things out a bit would be clearer.

Localize F^n by $\tau_R = \inf\{s : \int_0^s \|F_r^n\|_2^2 dr > R\}$ and $\hat{F}^{n,R} = F^n \mathbf{1}_{[0, \tau_R]}$. Let

$$X_R = S_t x + V_t + \int_0^t S_{t-s} \hat{F}_s^{n,R} ds.$$

Then (21) and (19) give

$$\text{Ent}(\text{Law}(X_R) \mid \nu_t^x) \leq \frac{1}{2} \mathbb{E} \int_0^t \|F_s^n\|_2^2 ds. \quad (22)$$

On $\{\tau_R > t\}$, (20) gives $X_R = u_t^n$, while $\mathbb{P}(\tau_R \leq t) \leq R^{-1} \mathbb{E} \int_0^t \|F_s^n\|_2^2 ds \rightarrow 0$. Hence $X_R \rightarrow u_t^n$ in probability in E . Lower semicontinuity of relative entropy on the Polish space E , then (19) and (22), yield

$$\sup_n \text{Ent}(P_t^n(x, \cdot) \mid \nu_t^x) < \infty. \quad (23)$$

By the assumed weak convergence (1) and lower semicontinuity again, $\text{Ent}(P_t(x, \cdot) \mid \nu_t^x) < \infty$. Thus $P_t(x, \cdot) \ll \nu_t^x$.

It remains only to compare Gaussian measures. In the basis $e_j(r) = \sqrt{2} \sin(j\pi r)$, $\lambda_j = j^2 \pi^2$, the covariance of ν_t^x has eigenvalues

$$q_j(t) = \frac{1 - e^{-\lambda_j t}}{\lambda_j},$$

whereas γ has eigenvalues λ_j^{-1} . The ratios $1 - e^{-\lambda_j t}$ are bounded away from zero and differ from 1 by a square-summable sequence. The Cameron–Martin spaces are therefore both $H_0^1(0, 1)$, and $S_t x \in H_0^1$. Feldman–Hajek gives $\nu_t^x \sim \gamma$. $\square$

Conclusion

Let η be an invariant probability measure. If $\pi(A) = 0$, then Lemma 3 gives $P_t(y, A) = 0$ for every $y \in E$, and hence

$$\eta(A) = \int_E P_t(y, A) \eta(dy) = 0.$$

Thus $\eta \ll \pi$; write $d\eta = h d\pi$. Detailed balance gives the L^1 -duality

$$\int f P_t g d\pi = \int P_t f g d\pi$$

for bounded Borel f and $g \in L^1(\pi)$, by truncating g . Therefore

$$\int f h d\pi = \int P_t f d\eta = \int P_t f h d\pi = \int f P_t h d\pi,$$

so $P_t h = h$ in $L^1(\pi)$. For $a > 0$, let $h_a = h \wedge a$. Since $r \mapsto r \wedge a$ is concave,

$$P_t h_a \leq (P_t h) \wedge a = h_a.$$

The two sides have the same π -integral; therefore $P_t h_a = h_a$. Now $h_a \in L^2(\pi)$, and the spectral gap (17) implies $\text{Var}_\pi(h_a) = 0$. Thus h_a is constant for every a . Letting $a \uparrow \infty$ and using $\int h d\pi = 1$, we obtain $h = 1$ π -a.s. Hence $\eta = \pi$.

Therefore P_t has at most one invariant probability measure on $C_0([0, 1])$. In fact, if an invariant probability exists, it is the Gibbs measure π in (13).

References

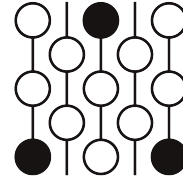
- [1] S. Athreya, O. Butkovsky, K. Lê and L. Mytnik, *Well-posedness of stochastic heat equation with distributional drift and skew stochastic heat equation*, Comm. Pure Appl. Math. 77 (2024), 2708–2777.
- [2] K. Lê, *Stochastic sewing in Banach spaces*, Electron. J. Probab. 28 (2023), paper no. 26, 1–22.
- [3] K. Lê, *A stochastic sewing lemma and applications*, Electron. J. Probab. 25 (2020), paper no. 38, 1–55.
- [4] V. I. Bogachev, *Gaussian Measures*, American Mathematical Society, 1998.
- [5] G. Da Prato and J. Zabczyk, *Ergodicity for Infinite Dimensional Systems*, Cambridge University Press, 1996.
- [6] Z.-M. Ma and M. Röckner, *Introduction to the Theory of (Non-Symmetric) Dirichlet Forms*, Springer, 1992.
- [7] S. K. Bounebacha and L. Zambotti, *A skew stochastic heat equation*, J. Theoret. Probab. 27 (2014), 168–201.

Review of Submission 5A

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

Correctness. After checking this proof carefully, I believe it to be correct. It handles the singularity of the delta function forcing by using the linear part of the equation to in some sense lower bound the size of the (space–time–probability) set at which the solution is equal to zero (although it is not phrased in exactly this way). This allows it to show that the contribution of the singular drift to the solution is in L^2 , so the Girsanov theorem can be used to show absolute continuity with respect to the Gaussian invariant measure. From here, standard techniques in infinite-dimensional ergodic theory allow it to show that the explicitly-constructed invariant measure is unique.

The approximation of the equation by its linearization on short time intervals introduces some error, which is bounded via the stochastic sewing lemma of Khoa Lê. There are some notational issues in the proofs that create some ambiguity and impede understanding to some degree but for an expert reader these issues are not serious.

Novelty. I rate the novelty of the solution as rather high. The use of stochastic sewing to control the difference between the solution to the linear and nonlinear problems on short time scales is very clean. To write such a solution, one would need to have expertise in ergodic theory, infinite-dimensional stochastic analysis, and the theory of rough stochastic equations. The proof actually proves a somewhat stronger statement than the human-generated solution (a finite-time absolute continuity result rather than the asymptotic strong Feller property).

Exposition The exposition could use some work. In particular, the importance of Lemma 1 should be signposted before it is proved. The reviewers agreed that it was easier to read the submission starting at the end and proceeding towards the beginning.

2 Recommendation

I rate the solution as essentially flawless. All of the mathematical arguments are present and are presented rather cleanly. An important issue that needs to be addressed, though, is that paper of Khoa Lê needs to be cited for the stochastic sewing lemma: in the submission, although the paper is mentioned in the bibliography, it is not cited in-text when the stochastic sewing lemma is presented.

A reader could be led to believe that the stochastic sewing lemma is due to this proposition, which is not the case.

3 Submission with inline comments

Uniqueness of the invariant measure

Problem statement and interpretation

Put $E = C_0([0, 1])$, $H = L^2(0, 1)$, and let S_t be the Dirichlet heat semigroup generated by $\frac{1}{2}\Delta$. I use the Dirichlet version of the Athreya–Butkovsky–Lê–Mytnik approximation assumption in the following form. For each deterministic initial condition $x \in E$, if $b_n = G_{1/n}$, then the regularized equation has the mild solution

$$u_t^n = S_t x + V_t + K_t^n, \quad K_t^n = \int_0^t S_{t-r} b_n(u_r^n) dr, \quad V_t = \int_0^t S_{t-r} dW_r,$$

and, for every $t > 0$ and every $\Phi \in C_b(E)$,

$$P_t^n \Phi(x) := \mathbb{E} \Phi(u_t^n) \longrightarrow P_t \Phi(x). \quad (1)$$

Thus I interpret the stated weak convergence of $u^n(t)$ to $u(t)$ as holding for all deterministic starting points; this is the form needed for a statement about invariant measures of the whole Markov semigroup.

Lemma 1 (Uniform estimate on the regularized drift). *For each $T < \infty$ there is $C_T < \infty$, independent of n and $x \in E$, such that*

$$\|K_t^n - S_{t-s} K_s^n\|_{L^2(\Omega; H)} \leq C_T |t - s|^{3/4}, \quad 0 \leq s < t \leq T. \quad (2)$$

Proof. Let p_t be the Dirichlet kernel and $d(y) = y \wedge (1 - y)$. For $0 < a \leq 2T$, the reflection formula for the interval and comparison with the two half-lines at the nearer boundary give

$$p_a(z, y) \leq C a^{-1/2} e^{-|z-y|^2/(Ca)}, \quad p_a(y, y) \asymp a^{-1/2} (1 \wedge d(y)^2/a). \quad (3)$$

For example, if $d(y) \leq \sqrt{a}$, the half-line expression $(2\pi a)^{-1/2} (1 - e^{-2d(y)^2/a})$ is comparable to $d(y)^2 a^{-3/2}$; if $d(y) \geq \sqrt{a}$, the reflected terms are exponentially smaller than the free diagonal term. The finitely many remaining images are absorbed in a constant depending only on T . Hence, for $\rho_\tau(y) = \int_0^\tau p_{2r}(y, y) dr$,

$$\rho_\tau(y) \asymp \sqrt{\tau} \wedge d(y), \quad \int_0^1 \rho_\tau(y)^{-1/2} dy \leq C_T \tau^{-1/4}. \quad (4)$$

Indeed the integral of $r^{-1/2} (1 \wedge d^2/r)$ is comparable to $\sqrt{\tau}$ when $d \geq \sqrt{\tau}$, and to d otherwise. Also

$$\sup_z \int_0^1 p_a(z, y) \rho_\tau(y)^{-1/2} dy \leq C_T (a^{-1/4} + \tau^{-1/4}). \quad (5)$$

To see this, use $\rho_\tau^{-1/2} \leq C(\tau^{-1/4} + d^{-1/2})$. The first part integrates to $C\tau^{-1/4}$. For the second, bound $d(y)^{-1/2} \leq y^{-1/2} + (1-y)^{-1/2}$ and use $y = \sqrt{a}r$ near the left boundary (similarly near the right) to get $Ca^{-1/4} \sup_{\alpha \geq 0} \int_0^\infty e^{-(r-\alpha)^2/C} r^{-1/2} dr < \infty$. Finally, for $0 < \tau \leq T$,

$$\|\rho_\tau^{-1} S_\tau f\|_2 \leq C_T \tau^{-1/2} \|f\|_2. \quad (6)$$

Here $\rho_\tau^{-1} \leq C(\tau^{-1/2} + d^{-1})$; $S_\tau f \in H_0^1$; Hardy's inequality gives $\|S_\tau f/d\|_2 \leq C\|\partial_x S_\tau f\|_2$; and $\|\partial_x S_\tau f\|_2 \leq C\tau^{-1/2} \|f\|_2$.

Write $D_{s,t}^n = K_t^n - S_{t-s} K_s^n$. Freeze the equation at time s :

$$A_{s,t}^n = \int_s^t S_{t-r} b_n(S_{r-s} u_s^n + V_r - S_{r-s} V_s) dr. \quad (7)$$

For fixed n , if Π is a partition of $[s, t]$, then

$$\sum_{[a,b] \in \Pi} S_{t-b} A_{a,b}^n \rightarrow D_{s,t}^n \quad \text{in } L^2(H). \quad (8)$$

Indeed the summand differs from $\int_a^b S_{b-r} b_n(u_r^n) dr$ by at most $C_n(b-a)^2$ in $L^2(H)$, because b_n is Lipschitz and the two arguments differ only by the drift accumulated on $[a, r]$, whose H -norm is at most $C_n(r-a)$. After applying S_{t-b} and summing, the error is $O_n((t-s)|\Pi|)$.

The estimates below are uniform in n . Given $\mathcal{F}_s$,

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The notation should indicate the dependence on s , e.g. $Z_{s,r}$.

$$Z_r(y) := S_{r-s} u_s^n(y) + V_r(y) - S_{r-s} V_s(y)$$

is Gaussian with variance $\rho_{r-s}(y)$. Since b_n is a probability density, $\sup_m \mathbb{E} b_n(m + N(0, \sigma^2)) \leq C\sigma^{-1}$. Moreover, given $\mathcal{F}_r$, the future part of $Z_q(z)$, $q > r$, has variance

$$\int_r^q \int_0^1 p_{q-\ell}(z, \xi)^2 d\xi d\ell = \int_0^{q-r} p_{2a}(z, z) da = \rho_{q-r}(z).$$

Thus, for $s < r < q$,

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The notation $\mathbb{E}_s$ needs to be defined as $\mathbb{E}[\cdot \mid \mathcal{F}_s]$ since this is not standard notation.

$$\mathbb{E}_s b_n(Z_r(y)) b_n(Z_q(z)) \leq C \rho_{r-s}(y)^{-1/2} \rho_{q-r}(z)^{-1/2}. \quad (9)$$

Using $\int p_{t-r}(x, y) p_{t-q}(x, z) dx = p_{2t-r-q}(y, z)$, then (5) in y and (4) in z , we obtain

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The notation $\|\cdot\|_2$ needs to be defined.

$$\mathbb{E}_s \|A_{s,t}^n\|_2^2 \leq C \int_s^t \int_r^t \left((2t-r-q)^{-1/4} + (r-s)^{-1/4} \right) (q-r)^{-1/4} dq dr \leq C|t-s|^{3/2}. \quad (10)$$

The last inequality is just scaling on the triangle $0 \leq r-s \leq q-s \leq t-s$. Hence $\|A_{s,t}^n\|_{L^2 H} \leq C|t-s|^{3/4}$.

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The statements in the following paragraph are vague. The reviewers wrote them out and believe that they are correct but they need to be explained more carefully.

For $s < u < t$, let $\delta A_{s,u,t}^n = A_{s,t}^n - S_{t-u}A_{s,u}^n - A_{u,t}^n$. The pieces before u cancel. For $r > u$, the two frozen arguments differ by $S_{r-u}D_{s,u}^n$. Conditional on $\mathcal{F}_u$, the remaining noise at space point y has variance $\rho_{r-u}(y)$, and $\|(b_n * \varphi_\sigma)'\|_\infty \leq C\sigma^{-2}$, uniformly in n . Therefore

$$|\mathbb{E}_u\{b_n(Z_r(y)) - b_n(Z_r(y) + h(y))\}| \leq C\rho_{r-u}(y)^{-1}|h(y)|.$$

With $h = S_{r-u}D_{s,u}^n$, (6) and contraction of S_t imply

$$\|\mathbb{E}_u\delta A_{s,u,t}^n\|_2 \leq C(t-u)^{1/2}\|D_{s,u}^n\|_2. \quad (11)$$

Taking $\mathbb{E}_s$ and using Jensen gives the same estimate with $\mathbb{E}_s$ in place of $\mathbb{E}_u$.

Reviewer 3

The above statement is difficult to interpret because we don't know what $\|\cdot\|_2$ is exactly supposed to mean. It is essentially true, however.

Reviewer 3

The following paragraph needs to cite the work of Khoa Lê referenced in the bibliography.

We need only the following dyadic sewing fact. Suppose $A_{s,t}$ is H -valued and $\mathcal{F}_t$ -measurable, and for every dyadic interval $[a, b]$ with midpoint u ,

$$\|\mathbb{E}_a\delta A_{a,u,b}\|_{L^2H} \leq \Gamma_1(b-a)^{1+\varepsilon_1}, \quad \|\delta A_{a,u,b} - \mathbb{E}_a\delta A_{a,u,b}\|_{L^2H} \leq \Gamma_2(b-a)^{1/2+\varepsilon_2}.$$

If $J_m = \sum_i S_{t-t_{i+1}}A_{t_i,t_{i+1}}$ over the 2^m equal subintervals of $[s, t]$, then J_m converges and

$$\|\mathcal{A}_{s,t} - A_{s,t}\|_{L^2H} \leq C(\Gamma_1|t-s|^{1+\varepsilon_1} + \Gamma_2|t-s|^{1/2+\varepsilon_2}). \quad (12)$$

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$\mathcal{A}_{s,t}$ needs to be defined.

Indeed, $J_{m+1} - J_m$ is the sum of $-S_{t-b}\delta A_{a,u,b}$. The conditional-mean parts are bounded by the triangle inequality, giving a factor $2^m(|t-s|2^{-m})^{1+\varepsilon_1}$. The centered parts are orthogonal: for two level- m intervals with the first lying earlier, the earlier summand is $\mathcal{F}_a$ -measurable for the later one, while the later summand has conditional mean zero given $\mathcal{F}_a$. Thus the square norm is the sum of the square norms. Summing the resulting geometric series proves (12).

For fixed n , $Y_n(h) = \sup_{0 < t-s \leq h} \|D_{s,t}^n\|_{L^2H}/|t-s|^{3/4}$ is finite, because b_n is bounded. On intervals of length at most h , (11) gives

$$\|\mathbb{E}_a\delta A_{a,u,b}^n\|_{L^2H} \leq CY_n(h)(b-a)^{5/4}.$$

Also

$$\|\delta A_{a,u,b}^n - \mathbb{E}_a\delta A_{a,u,b}^n\|_{L^2H} \leq 2\|\delta A_{a,u,b}^n\|_{L^2H} \leq C(b-a)^{3/4},$$

because δA is a sum of three A -terms and (10) applies to each. The dyadic sewing limit is $D_{s,t}^n$ by (8).

The explicit limit should be written out instead of calling it the “dyadic sewing limit,” which is vague.

Combining (10) and (12),

$$\|D_{s,t}^n\|_{L^2H} \leq C|t-s|^{3/4} + CY_n(h)|t-s|^{5/4}.$$

Thus $Y_n(h) \leq C + Ch^{1/2}Y_n(h)$. Choose $h_0 > 0$, depending only on T , so that $Ch_0^{1/2} \leq 1/2$; then $Y_n(h_0) \leq 2C$, uniformly in n . For longer intervals use the additivity $D_{s,t}^n = S_{t-r}D_{s,r}^n + D_{r,t}^n$ and split into pieces of length at most h_0 . This gives a linear bound $C_T|t-s|$, which is bounded by $C_T|t-s|^{3/4}$ on $0 \leq t-s \leq T$. The lemma follows. $\square$

The reversible reference measure and its gap

Let γ be Brownian bridge measure on E , the centered Gaussian measure with covariance $(-\Delta_D)^{-1}$. Define

$$\Lambda(v) = \int_0^1 \mathbf{1}_{\{v(r)>0\}} dr, \quad \pi(dv) = Z^{-1} e^{2\Lambda(v)} \gamma(dv). \quad (13)$$

For $B'_n = b_n$, $B_n(a) = \int_{-\infty}^a b_n(r) dr$, put

$$\pi_n(dv) = Z_n^{-1} \exp\left(2 \int_0^1 B_n(v(r)) dr\right) \gamma(dv).$$

Since $0 \leq B_n \leq 1$, $B_n(a) \rightarrow \mathbf{1}_{\{a>0\}}$ for $a \neq 0$, and a Brownian bridge spends zero Lebesgue time at a fixed level,

$$\|\pi_n - \pi\|_{TV} \rightarrow 0. \quad (14)$$

All densities $d\pi_n/d\gamma$ and $d\pi/d\gamma$ are bounded above and below by deterministic positive constants.

Lemma 2. *There are constants $C, c > 0$, independent of n , such that for every bounded continuous f, g and every $t \geq 0$,*

$$\int P_t^n f d\pi_n = \int f d\pi_n, \quad \int g P_t^n f d\pi_n = \int f P_t^n g d\pi_n, \quad (15)$$

and

$$\text{Var}_{\pi_n}(P_t^n f) \leq e^{-ct} \text{Var}_{\pi_n}(f). \quad (16)$$

Proof. Let Π_m be projection onto the first m Dirichlet modes and set

$$U_{n,m}(v) = \int_0^1 B_n(\Pi_m v(r)) dr.$$

Its H -gradient is $\nabla_H U_{n,m}(v) = \Pi_m b_n(\Pi_m v)$. Let $P_t^{n,m}$ be the semigroup of

$$u_t^{n,m} = S_t x + V_t + \int_0^t S_{t-r} \Pi_m b_n(\Pi_m u_r^{n,m}) dr.$$

The first m Fourier coordinates solve the finite-dimensional Langevin diffusion

$$dX_j = -(\lambda_j/2)X_j dt + \partial_j U_{n,m}(X) dt + d\beta_j, \quad j \leq m,$$

and the tail is an independent OU process. Hence, by finite-dimensional integration by parts and the standard OU integration by parts in the tail, $P_t^{n,m}$ is reversible with respect to $\pi_{n,m} = Z_{n,m}^{-1} e^{2U_{n,m}\gamma}$, and its Dirichlet form is $\mathcal{E}_{n,m}(F, F) = \frac{1}{2} \int \|\nabla_H F\|_2^2 d\pi_{n,m}$ on smooth cylinder functions.

The measures $\pi_{n,m}$ converge to π_n in total variation: $\Pi_m v \rightarrow v$ in H , γ -a.s., and B_n is bounded and Lipschitz. For fixed n , $u_t^{n,m} \rightarrow u_t^n$ in probability in E . In H , this follows from Gronwall, since $\Pi_m b_n(\Pi_m z)$ is uniformly Lipschitz in z for fixed n , and for each H -valued y , $\Pi_m b_n(\Pi_m y) \rightarrow b_n(y)$ in H , dominated by $2\|b_n\|_\infty$. For convergence in E at a fixed $t > 0$, split the drift convolution into $[0, t - \varepsilon]$ and $[t - \varepsilon, t]$. On the first part, $S_{t-r} : H \rightarrow H_0^\alpha \subset E$, $\alpha > 1/2$, is bounded. On the short terminal part,

$$\int_{t-\varepsilon}^t \|S_{t-r} F_r\|_\infty dr \leq C \|b_n\|_\infty \int_0^\varepsilon a^{-1/4} da,$$

for both $F_r = \Pi_m b_n(\Pi_m u_r^{n,m})$ and $F_r = b_n(u_r^n)$, because Π_m is an L^2 -contraction and $S_a : H \rightarrow E$ has norm $C a^{-1/4}$. This tends to zero with ε , uniformly in m .

Consequently the reversible identities pass to $m = \infty$. For instance, if $h_m = d\pi_{n,m}/d\gamma$, then $h_m \rightarrow h_n = d\pi_n/d\gamma$ in $L^1(\gamma)$ and the h_m 's are uniformly bounded; together with $P_t^{n,m} f(x) \rightarrow P_t^n f(x)$ and bounded convergence this gives (15).

It remains to record the gap uniformly. The Gaussian Poincare inequality for γ in H -gradient form is

$$\text{Var}_\gamma(F) \leq \lambda_1^{-1} \int \|\nabla_H F\|_2^2 d\gamma$$

on smooth cylinder functions. Since all densities $d\pi_{n,m}/d\gamma$ lie in a fixed interval $[a_0, a_0^{-1}]$, the Holley–Stroock bounded perturbation argument gives

$$\text{Var}_{\pi_{n,m}}(F) \leq C \mathcal{E}_{n,m}(F, F),$$

with C independent of n, m . For a symmetric Markov semigroup this Poincare inequality is equivalent to exponential L^2 -decay, so

$$\text{Var}_{\pi_{n,m}}(P_t^{n,m} f) \leq e^{-ct} \text{Var}_{\pi_{n,m}}(f).$$

Letting $m \rightarrow \infty$, using the convergence just proved and the uniformly bounded densities, yields (16). □

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The following proposition should be cited to the work of Bounebaché–Zambotti (2014).

Proposition 1. *The measure π is P_t -invariant, P_t is symmetric on $L^2(\pi)$, and*

$$\text{Var}_\pi(P_t f) \leq e^{-ct} \text{Var}_\pi(f), \quad f \in L^2(\pi), \quad t \geq 0. \quad (17)$$

Proof. For $f, g \in C_b(E)$, (1), (14), (15), and the uniform boundedness of the densities imply, by dominated convergence with respect to γ ,

$$\int P_t f d\pi = \lim_n \int P_t^n f d\pi_n = \int f d\pi, \quad \int g P_t f d\pi = \int f P_t g d\pi.$$

On the Polish space E , these identities identify the measure $\int \pi(dx) P_t(x, \cdot)$ with π , and the measure $\pi(dx) P_t(x, dy)$ with its transpose; hence invariance and detailed balance hold for all bounded Borel functions. Similarly, (16) passes to the limit for bounded continuous f . Finally $C_b(E)$ is dense in $L^2(\pi)$, and an invariant Markov operator is an L^2 -contraction, so symmetry and (17) extend to all $L^2(\pi)$. $\square$

Fixed-time absolute continuity

Lemma 3. *For every deterministic $x \in E$ and every $t > 0$,*

$$P_t(x, \cdot) \ll \gamma,$$

and hence $P_t(x, \cdot) \ll \pi$.

Proof. Let $\nu_t^x = \text{Law}(S_t x + V_t)$. We first bound $\text{Ent}(P_t^n(x, \cdot) \mid \nu_t^x)$ uniformly in n . Put $r_k = t(1 - 2^{-k})$, $\Delta_k = r_{k+1} - r_k$, and $D_k^n = K_{r_{k+1}}^n - S_{\Delta_k} K_{r_k}^n$. Define

$$F_s^n = \sum_{k \geq 0} \frac{\mathbf{1}_{(r_{k+1}, r_{k+2}]}(s)}{\Delta_{k+1}} S_{s-r_{k+1}} D_k^n. \quad (18)$$

The process is predictable because D_k^n is $\mathcal{F}_{r_{k+1}}$ -measurable. Lemma 1 and contraction of S_t give

$$\mathbb{E} \int_0^t \|F_s^n\|_2^2 ds \leq C_t \sum_{k \geq 0} \frac{\Delta_k^{3/2}}{\Delta_{k+1}} < \infty, \quad (19)$$

uniformly in n . For partial sums,

$$\int_0^t S_{t-s} F_s^{n,N} ds = \sum_{k=0}^N S_{t-r_{k+1}} D_k^n = S_{t-r_{N+1}} K_{r_{N+1}}^n,$$

where $F^{n,N}$ denotes the sum in (18) up to $k = N$. Since $r_{N+1} \uparrow t$, the regularized mild solution gives K^n continuous with values in E , and $S_a \rightarrow I$ strongly on E ,

$$\int_0^t S_{t-s} F_s^n ds = K_t^n \quad (20)$$

in probability in E (and hence in H). The state space assertion is legitimate because, for every deterministic $F \in L^2(0, t; H)$,

$$\sum_j \lambda_j^\alpha \left| \int_0^t e^{-\lambda_j(t-s)/2} F_j(s) ds \right|^2 \leq \sum_j \lambda_j^{\alpha-1} \int_0^t |F_j(s)|^2 ds \leq C \|F\|_{L^2(0,t;H)}^2$$

for $0 < \alpha < 1$; choosing $\alpha > 1/2$ gives an element of $H_0^\alpha(0, 1) \subset E$.

We use the following standard entropy form of Girsanov, and recall its proof. If F is predictable, H -valued, and has bounded energy, then

$$X^F = S_t x + V_t + \int_0^t S_{t-s} F_s ds$$

satisfies

$$\text{Ent}(\text{Law}(X^F) \mid \nu_t^x) \leq \frac{1}{2} \mathbb{E} \int_0^t \|F_s\|_2^2 ds. \quad (21)$$

For the first M Brownian coordinates this is finite-dimensional Girsanov: under the tilted measure, the shifted coordinates $\beta_j + \int_0^\cdot F_j(s) ds$, $j \leq M$, are Brownian, and the entropy cost is $\frac{1}{2} \mathbb{E} \int_0^t \sum_{j \leq M} F_j(s)^2 ds$.

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Should say “applying the data processing inequality”

Applying data processing to the deterministic-kernel stochastic convolution gives the same bound for the first M terminal Fourier coordinates. Since the Borel sigma-field of E is generated by these coordinates and relative entropy is the supremum over increasing finite-coordinate sigma-fields, (21) follows.

Localize F^n by $\tau_R = \inf\{s : \int_0^s \|F_r^n\|_2^2 dr > R\}$ and $\hat{F}^{n,R} = F^n \mathbf{1}_{[0, \tau_R]}$. Let

$$X_R = S_t x + V_t + \int_0^t S_{t-s} \hat{F}_s^{n,R} ds.$$

Then (21) and (19) give

$$\text{Ent}(\text{Law}(X_R) \mid \nu_t^x) \leq \frac{1}{2} \mathbb{E} \int_0^t \|F_s^n\|_2^2 ds. \quad (22)$$

On $\{\tau_R > t\}$, (20) gives $X_R = u_t^n$, while $\mathbb{P}(\tau_R \leq t) \leq R^{-1} \mathbb{E} \int_0^t \|F_s^n\|_2^2 ds \rightarrow 0$. Hence $X_R \rightarrow u_t^n$ in probability in E . Lower semicontinuity of relative entropy on the Polish space E , then (19) and (22), yield

$$\sup_n \text{Ent}(P_t^n(x, \cdot) \mid \nu_t^x) < \infty. \quad (23)$$

By the assumed weak convergence (1) and lower semicontinuity again, $\text{Ent}(P_t(x, \cdot) \mid \nu_t^x) < \infty$. Thus $P_t(x, \cdot) \ll \nu_t^x$.

It remains only to compare Gaussian measures. In the basis $e_j(r) = \sqrt{2} \sin(j\pi r)$, $\lambda_j = j^2 \pi^2$, the covariance of ν_t^x has eigenvalues

$$q_j(t) = \frac{1 - e^{-\lambda_j t}}{\lambda_j},$$

whereas γ has eigenvalues λ_j^{-1} . The ratios $1 - e^{-\lambda_j t}$ are bounded away from zero and differ from 1 by a square-summable sequence. The Cameron–Martin spaces are therefore both $H_0^1(0, 1)$, and $S_t x \in H_0^1$. Feldman–Hajek gives $\nu_t^x \sim \gamma$. $\square$

Conclusion

Let η be an invariant probability measure. If $\pi(A) = 0$, then Lemma 3 gives $P_t(y, A) = 0$ for every $y \in E$, and hence

$$\eta(A) = \int_E P_t(y, A) \eta(dy) = 0.$$

Thus $\eta \ll \pi$; write $d\eta = h d\pi$. Detailed balance gives the L^1 -duality

$$\int f P_t g d\pi = \int P_t f g d\pi$$

for bounded Borel f and $g \in L^1(\pi)$, by truncating g . Therefore

$$\int f h d\pi = \int P_t f d\eta = \int P_t f h d\pi = \int f P_t h d\pi,$$

so $P_t h = h$ in $L^1(\pi)$. For $a > 0$, let $h_a = h \wedge a$. Since $r \mapsto r \wedge a$ is concave,

$$P_t h_a \leq (P_t h) \wedge a = h_a.$$

The two sides have the same π -integral; therefore $P_t h_a = h_a$. Now $h_a \in L^2(\pi)$, and the spectral gap (17) implies $\text{Var}_\pi(h_a) = 0$. Thus h_a is constant for every a . Letting $a \uparrow \infty$ and using $\int h d\pi = 1$, we obtain $h = 1$ π -a.s. Hence $\eta = \pi$.

Therefore P_t has at most one invariant probability measure on $C_0([0, 1])$. In fact, if an invariant probability exists, it is the Gibbs measure π in (13).

References

- [1] S. Athreya, O. Butkovsky, K. Lê and L. Mytnik, *Well-posedness of stochastic heat equation with distributional drift and skew stochastic heat equation*, Comm. Pure Appl. Math. 77 (2024), 2708–2777.
- [2] K. Lê, *Stochastic sewing in Banach spaces*, Electron. J. Probab. 28 (2023), paper no. 26, 1–22.
- [3] K. Lê, *A stochastic sewing lemma and applications*, Electron. J. Probab. 25 (2020), paper no. 38, 1–55.
- [4] V. I. Bogachev, *Gaussian Measures*, American Mathematical Society, 1998.
- [5] G. Da Prato and J. Zabczyk, *Ergodicity for Infinite Dimensional Systems*, Cambridge University Press, 1996.
- [6] Z.-M. Ma and M. Röckner, *Introduction to the Theory of (Non-Symmetric) Dirichlet Forms*, Springer, 1992.
- [7] S. K. Bounebacha and L. Zambotti, *A skew stochastic heat equation*, J. Theoret. Probab. 27 (2014), 168–201.